2024-3-60-082 PHY-209 . Section-3

Assignment

1| a) For a normalized wave function,

$$\int_{-\infty}^{+\infty} |\psi|^2 \, dx = 1$$

$$\therefore \int_0^1 (ax)^2 \, dx = 1$$

$$\Rightarrow a^2 \int_0^1 x^2 \, dx = 1$$

$$\Rightarrow a^2 \left[\frac{x^3}{3}\right]_0^1 = 1$$

$$\Rightarrow a^2 \left[\frac{1^3}{3} - \frac{0^3}{3}\right] = 1$$

$$\Rightarrow \frac{a^2}{3} = 1$$

$$\Rightarrow a = \sqrt{3}$$ Ans

b) $$P = \int_{0.45}^{0.55} |\psi|^2 \, dx$$

$$= \int_{0.45}^{0.55} (\sqrt{3}\,x)^2 \, dx$$

$$= 3 \int_{0.45}^{0.55} x^2 \, dx$$

$$= 3 \left[\frac{x^3}{3}\right]_{0.45}^{0.55} \cancel{dx}$$

$$= 3 \left[\frac{(0.55)^3}{3} - \frac{(0.45)^3}{3}\right]$$

$$= 0.075$$

$$= 7.5\ \%$$

c)

$$\langle x \rangle = \int_0^1 x\, |\psi|^2\, dx$$

$$= \int_0^1 x\, (\sqrt{3}\, x)^2\, dx$$

$$= \int_0^1 x \times 3x^2\, dx$$

$$= 3 \int_0^1 x^3\, dx$$

$$= 3 \left[\frac{x^4}{4} \right]_0^1$$

$$= 3 \times \left[\frac{1}{4} - \frac{0}{4} \right]$$

$$= \frac{3}{4}$$

$$= 0.75$$

Ans

21.a) Given, $\psi(x) = a(1-x)$

Probability between 0.3 to 0.6:

$$P = \int_{0.3}^{0.6} |\psi|^2 \, dx$$

Now, for a normalized wave function,

$$\int_0^1 |\psi|^2 \, dx = 1$$

$$\Rightarrow \int_0^1 (a(1-x))^2 \, dx = 1$$

$$\Rightarrow a^2 \int_0^1 (1-2x+x^2) \, dx = 1$$

$$\Rightarrow a^2 \int_0^1 1 \, dx - \int_0^1 2x \, dx + \int_0^1 x^2 \, dx = 1$$

$$\Rightarrow a^2 \left[x - \frac{2x^2}{2} + \frac{x^3}{3}\right]_0^1 = 1$$

$$\Rightarrow a^2 \left[1 - 1^2 + \frac{1^3}{3}\right] = 1$$

$$\Rightarrow \frac{a^2}{3} = 1$$

$$\therefore a = \sqrt{3}$$

$$\therefore \psi(x) = \sqrt{3}(1-x)$$

$$\therefore P = \int_{0.3}^{0.6} |\psi|^2 \, dx$$

$$\Rightarrow P = \int_{0.3}^{0.6} (\sqrt{3}(1-x))^2 \, dx$$

$$\Rightarrow P = 3\int_{0.3}^{0.6} (1-2x+x^2) \, dx$$

$$\Rightarrow P = 3\left[x - x^2 + \frac{x^3}{3}\right]_{0.3}^{0.6}$$

$$\Rightarrow P = 3\left[\frac{39}{125} - \frac{219}{1000}\right]$$

$$\Rightarrow P = 0.279$$

$$\therefore P = 27.9\ \%$$

b) $$\langle x \rangle = \int_0^1 x|\psi|^2\, dx$$

$$= \int_0^1 x \cdot 3 \cdot (1-x)^2\, dx$$

$$= 3\int_0^1 x(1-2x+x^2)\, dx$$

$$= 3\int_0^1 x - 2x^2 + x^3\, dx$$

$$= 3\left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right]_0^1$$

$$= 3 \times \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right)$$

$$= 0.25$$

Ans

31 Given, $\psi(x) = A\sin 2x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Normalization :

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\psi|^2 \, dx = 1$$

$$\Rightarrow A^2 \int_{-\pi/2}^{\pi/2} \sin^2 2x \, dx = 1$$

$$\Rightarrow \frac{A^2}{2} \int_{-\pi/2}^{\pi/2} 1 - \cos 4x \, dx = 1$$

$$\Rightarrow \frac{A^2}{2} \left[x - \frac{\sin 4x}{4} \right]_{-\pi/2}^{\pi/2} = 1$$

$$\Rightarrow \frac{A^2}{2} \left(\pi/2 - \frac{\sin 2\pi}{4} \right) + \left(\frac{\pi}{2} + \frac{\sin(-2\pi)}{4} \right) = 1$$

$$\Rightarrow \frac{A^2}{2} \left(\frac{2\pi}{2} \right) = 1$$

$$\Rightarrow \frac{A^2}{2} \times \pi = 1$$

$$\Rightarrow A^2 = \frac{2}{\pi}$$

$$\Rightarrow A = \sqrt{\frac{2}{\pi}}$$

$$\therefore \psi = \sqrt{\frac{2}{\pi}} \sin 2x$$

Now, $P = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} |\psi|^2 \, dx$

$$= \int_{-\pi/4}^{\pi/4} A^2 \sin^2 2x \, dx$$

$$= \int_{-\pi/4}^{\pi/4} A^2 \, \frac{1-\cos 4x}{2} \, dx$$

$$= \frac{1}{\pi} \int_{-\pi/4}^{\pi/4} (1-\cos 4x) \, dx$$

$$= \frac{1}{\pi} \left[x - \frac{\sin 4x}{4} \right]_{-\pi/4}^{\pi/4}$$

$$= \frac{1}{\pi} \left[\left(\frac{\pi}{4} - \frac{\sin \pi}{4} \right) - \left(-\frac{\pi}{4} - \frac{\sin(-\pi)}{4} \right) \right]$$

$$= \frac{1}{\pi} \left(\frac{\pi}{4} + \frac{\pi}{4} \right)$$

$$= \frac{1}{\pi} \times \frac{\pi}{2}$$

$$= \frac{1}{2}$$

$$= 0.5$$

$$= 50\%$$

4]

Given, Operator, $\hat{O} = \frac{d^2}{dx^2}$

$\psi = \sin nx$

Let, Eigen Value $= \lambda$

$\therefore \hat{O}\psi = \lambda\psi$

Now, $\frac{d\psi}{dx} = \frac{d}{dx}\sin(nx)$

$= n\cos(nx)$

and, $\frac{d^2\psi}{dx^2} = -n^2\sin(nx)$

$\Rightarrow \frac{d^2\psi}{dx^2} = -n^2\psi$

$\therefore$ Eigen value, $\lambda = -n^2$ $[n = 1, 2, 3, \ldots]$

corresponding eigenvalues $= -1, -4, -9, \ldots$

5| Given, $\psi = Ae^{i(kx-\omega t)}$

Momentum, $\hat{p}\psi = -i\hbar \frac{\partial \psi}{\partial x}$

$= -i\hbar \frac{\partial}{\partial x} Ae^{i(kx-wt)}$

$= -i\hbar Ae^{i(kx-w)} \frac{\partial}{\partial x} i(kx-wt)$

$= -i\hbar \cdot \psi \cdot ik$

$= \hbar k \psi$

$\therefore$ Momentum Eigenvalue, $p = \hbar k$

Energy, $\hat{E}\psi = [i\hbar \frac{\partial}{\partial x} Ae^{i(kx-wt)}]$

$= i\hbar Ae^{i(kx-wt)} \frac{\partial}{\partial x} (i(kx-wt))$

$= i\hbar\psi(-iw)$

$= \hbar w \psi$

$\therefore$ Energy Eigenvalue, $E = \hbar w$

6)

wave function of a particle trapped in a box L :

$$\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

$$\therefore P = \int_{0.45L}^{0.55L} \left(\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}\right)^2 dx$$

$$= \frac{2}{L} \int_{0.45L}^{0.55L} \sin^2 \frac{n\pi x}{L}\, dx$$

$$= \frac{2}{L} \int_{0.45L}^{0.55L} \frac{1-\cos \frac{2n\pi x}{L}}{2}\, dx$$

$$= \frac{1}{L} \int_{0.45L}^{0.55L} 1-\cos \frac{2n\pi x}{L}\, dx$$

$$= \frac{1}{L} \left[x - \frac{\sin \frac{2n\pi x}{L}}{\frac{2\pi n}{L}} \right]_{0.45L}^{0.55L}$$

$$= \frac{1}{L} \left[(0.55L - 0.45L) - \frac{L}{2\pi n} \left(\sin \frac{2\pi n \times 0.55L}{L} - \sin \frac{2\pi n \times 0.45L}{L} \right) \right.$$

$$= \left[0.10 - \frac{1}{2n\pi} \left(\sin 1.1\pi n - \sin 0.9 n\pi \right) \right]$$

Ground State:

for $n = 1$, $P = 0.10 - \frac{1}{2\pi} (\sin 1.1\pi - \sin 0.9\pi)$

$= 0.198 \quad = 19.83\%$

First Excited State:

for $n = 2$, $P = 0.10 - \frac{1}{2 \times 2\pi} (\sin 1.1 \times 2\pi - \sin 0.9 \times 2\pi)$

$= 6.45 \times 10^{-3} \quad = 0.645\%$

71 $\langle x \rangle = \int_{-\infty}^{+\infty} x |\psi|^2 dx$

$$= \int_0^L x \left(\sqrt{\tfrac{2}{L}} \sin \frac{n\pi x}{L}\right)^2 dx$$

$$= \frac{2}{L} \int_0^L x \sin^2 \frac{n\pi x}{L} \, dx$$

$$= \frac{2}{L} \left[\frac{x^2}{4} - \frac{x \sin \frac{2n\pi x}{L}}{\frac{4n\pi}{L}} - \frac{\cos \frac{2n\pi x}{L}}{8\left(\frac{n\pi}{L}\right)^2} \right]_0^L$$

$$= \frac{2}{L} \times \frac{L^2}{4}$$

$$= \frac{L}{2}$$

81 Given, $|\psi_1\rangle = (1+i)|\phi_1\rangle + 2|\phi_2\rangle + b|\phi_3\rangle + 3i|\phi_4\rangle$

$|\psi_2\rangle = 2|\phi_1\rangle + i|\phi_2\rangle - 4|\phi_3\rangle + |\phi_4\rangle$

$|\psi_1\rangle$ and $|\psi_2\rangle$ would be orthogonal if there inner product is 0.

$\therefore \langle\psi_1|\psi_2\rangle = 0$

Now, $|\psi_1\rangle = \begin{bmatrix} 1+i \\ -2 \\ b \\ 3i \end{bmatrix}$ $\quad |\psi_2\rangle = \begin{bmatrix} 2 \\ i \\ -4 \\ 1 \end{bmatrix}$

$\therefore \langle\psi_1| = [1-i \quad -2 \quad b^* \quad -3]$ $\quad \therefore \langle\psi_2| = [2 \quad -i \quad -4 \quad 1]$

Now, $\langle\psi_1|\psi_2\rangle = 0$

$\Rightarrow [1-i \quad -2 \quad b^* \quad -3i] \begin{bmatrix} 2 \\ i \\ -4 \\ 1 \end{bmatrix} = 0$

$\Rightarrow 2-2i-2i-4b^*-3i = 0$

$\Rightarrow 2-7i-4b^* = 0$

$\therefore b^* = \dfrac{2-7i}{4}$

$\therefore b = \dfrac{2+7i}{4}$

9)

Given, $|\psi\rangle = 3i|\phi_1\rangle - 7i|\phi_2\rangle$

$|\chi\rangle = -|\phi_1\rangle + 2i|\phi_2\rangle$

$\therefore |\psi\rangle = \begin{bmatrix} 3i \\ -7i \end{bmatrix}$ $\therefore |\chi\rangle = \begin{bmatrix} -1 \\ 2i \end{bmatrix}$

a) $|\psi + \chi\rangle = \begin{bmatrix} 3i - 1 \\ -7i + 2i \end{bmatrix}$

$= \begin{bmatrix} 3i - 1 \\ -5i \end{bmatrix}$

$\langle\psi| = \begin{bmatrix} -3i & 7i \end{bmatrix}$ $\langle\chi| = \begin{bmatrix} -1 & -2i \end{bmatrix}$

$\therefore \langle\psi + \chi| = \begin{bmatrix} -3i - 1 & 7i - 2i \end{bmatrix}$

$= \begin{bmatrix} -3i - 1 & 5i \end{bmatrix}$

b) L.H.S.

$\langle\psi|\chi\rangle$

$= \begin{bmatrix} -3i & 7i \end{bmatrix} \begin{bmatrix} -1 \\ 2i \end{bmatrix}$

$= 3i + 14i^2$

$= 3i - 14$

R.H.S.

$\langle\chi|\psi\rangle$

$= \begin{bmatrix} -1 & -2i \end{bmatrix} \begin{bmatrix} 3i \\ -7i \end{bmatrix}$

$= \begin{bmatrix} -3i + 14i^2 \end{bmatrix}$

$= -14 - 3i$

$\therefore \langle\psi|\chi\rangle \neq \langle\chi|\psi\rangle$

10)

Given,

$$|\psi\rangle = \begin{bmatrix} -3i \\ 2+i \\ 4 \end{bmatrix} \qquad |\phi\rangle = \begin{bmatrix} 2 \\ -i \\ 2-3i \end{bmatrix}$$

a) $\langle \phi | = \begin{bmatrix} 2 & i & 2+3i \end{bmatrix}$ Ans

b) $\langle \phi | = \begin{bmatrix} 2 & i & 2+3i \end{bmatrix}$

$\langle \psi | = \begin{bmatrix} 3i & 2-i & 4 \end{bmatrix}$

$$\therefore \langle \phi | \psi \rangle = \begin{bmatrix} 2 & i & 2+3i \end{bmatrix} \begin{bmatrix} -3i \\ 2+i \\ 4 \end{bmatrix}$$

$$= -6i + 2i + i^2 + 8 + 12i$$

$$= 7 + 8i \quad \text{Ans}$$

c) $|\psi\rangle|\phi\rangle$ is a product of 3×1 and 3×1 matrix. which is not permissible. Column of first matrix and row of second matrix should be equal for multiplication.

Also, $\langle \phi | \langle \psi |$ is a product of 1×3 and 1×3 matrix. which is not permisible. Column of first matrix and row of second matrix should be equal for multiplication.

11)

We know, Momentum Operator, $\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$

Hamiltonian Operator, $\hat{H} = \frac{\hat{p}^2}{2m} + V$

Now, $[\hat{p}, \hat{H}] = \hat{p}\hat{H} - \hat{H}\hat{p}$

$= \hat{p}\left(\frac{\hat{p}^2}{2m} + V\right) - \left(\frac{\hat{p}^2}{2m} + V\right)\hat{p}$

$= \frac{\hat{p}^3}{2m} + \hat{p}V - \frac{\hat{p}^3}{2m} + \hat{p}V$

$= 0$

So, the momentum operator and Hamiltonian operator commut when the potential energy is constant.

|2|

Let, Arbitary function, $f(x)$

Now, $[\hat{x}, \frac{\partial}{\partial x}] f(x) = \hat{x}\left(\frac{\partial}{\partial x} f(x)\right) - \frac{\partial}{\partial x}\left(\hat{x} f(x)\right)$

$\Rightarrow [\hat{x}, \frac{\partial}{\partial x}] f(x) = \hat{x} f'(x) - x f'(x) - f(x)$

$\Rightarrow [\hat{x}, \frac{\partial}{\partial x}] f(x) = - f(x)$

$\therefore [\hat{x}, \frac{\partial}{\partial x}] = -1$

Proved

|3|

Let, Arbitary function $f(x)$

L.H.S.

$\hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} f(x)$

$= \hat{x}^2 \frac{\partial}{\partial x} \cdot \frac{f(x)}{\hat{x}}$

$= \hat{x}^2 \frac{\hat{x} f'(x) - f(x)}{\hat{x}^2}$

$= x f'(x) - f(x)$

$\therefore \hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} f(x) = x f'(x) - f(x) = x \frac{d}{dx} f(x) - f(x)$

$\Rightarrow \hat{x}^2 \frac{d}{dx} \frac{1}{\hat{x}} = x\frac{d}{dx} - 1$

14) We know, $\hat{p} = -i\hbar \frac{\partial}{\partial x}$

Let, arbitarty function, $f(x)$

$\therefore [\hat{P}_x, \hat{V}(x)] f(x)$

$= \hat{P}_x (V(x) f(x)) - \hat{V}(x) (\hat{P}_x f(x))$

$= -i\hbar \frac{\partial}{\partial x} (V(x) f(x)) - \hat{V}(x) (-i\hbar \frac{\partial}{\partial x} f(x))$

$= -i\hbar [\hat{V}(x) f'(x) + f(x) \hat{V}'(x)] + \hat{V}(x) i\hbar f'(x)$

$= -i\hbar \hat{V}(x) f'(x) - i\hbar f(x) \hat{V}'(x) + i\hbar \hat{V}(x) f'(x)$

$= -i\hbar f(x) \frac{\partial V(x)}{\partial x}$

Showed

15| Given, $|\alpha\rangle = \begin{bmatrix} 2i \\ -1 \\ 3 \end{bmatrix}$ $|\beta\rangle = \begin{bmatrix} 4 \\ -2i \\ i \end{bmatrix}$

$\therefore \langle\alpha| = \begin{bmatrix} -2i & -1 & 3 \end{bmatrix}$

$|\alpha\rangle$ and $|\beta\rangle$ will be orthogonal if $\langle\alpha|\beta\rangle = 0$ satisfy.

$\therefore \langle\alpha|\beta\rangle = \begin{bmatrix} -2i & -1 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ -2i \\ i \end{bmatrix}$

$= -8i + 2i + 3i$

$= -3i \neq 0$

$\therefore$ $|\alpha\rangle$ and $|\beta\rangle$ are not orthogonal.

consider $2i$ as x

$\therefore |\alpha\rangle = \begin{bmatrix} x \\ -1 \\ 3 \end{bmatrix}$ $\therefore \langle\alpha| = \begin{bmatrix} x^* & -1 & 3 \end{bmatrix}$

Now, $\langle\alpha|\beta\rangle = 0$

$\Rightarrow \begin{bmatrix} x^* & -1 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ -2i \\ i \end{bmatrix} = 0$

$\Rightarrow 4x^* + 2i + 3i = 0$

$\Rightarrow 4x^* = -5i$

$\Rightarrow x^* = -\frac{5}{4}i$

$\therefore x = \frac{5}{4}i$

$\therefore$ if $|\alpha\rangle = \begin{bmatrix} \frac{5}{4}i \\ -1 \\ 3 \end{bmatrix}$ and $|\beta\rangle = \begin{bmatrix} 4 \\ -2i \\ i \end{bmatrix}$ then they will ~~be~~come orthogonal.